MMA3001 Project

Limited-Observation Calibration of Composite Beam Compliance Models

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24 October 2026

Abstract

This project demonstrates a workflow for fitting a finite-element-analysis (FEA) response model, calibrating it with a small number of observations, and testing predictions on withheld cases. The updated PyAnsys MAPDL model uses large-deflection static analysis for 80 training cases and 100 target cases across five sets. The fifth set adds a doubled resin-to-fibre ratio as an unmodelled composition shift. Three model classes are compared: a mechanics-informed compliance regression, a quadratic response surface, and a Gaussian process. Each target set uses three repeats of two-observation carbon/glass calibration with 18 held-out cases. The 180-case nonlinear FEA sweep has not yet been run; numeric validation results in this report are from the earlier linear-static, four-set baseline and must not be attributed to the updated experiment. The historical trial showed that two observations do not always identify two material corrections. The alternate-layup baseline used known configured material cards, not a blind unknown-material prediction. This study is FEA-to-FEA only and does not validate physical material properties or a curved landing-gear leg.

1 Introduction

Composite landing-gear design can require costly physical testing to determine how geometry, layup, and laminate stiffness affect deflection. This project investigates whether a model trained on FEA results can be corrected with only a few observations and used to predict unobserved responses. The long-term motivation is to apply the same workflow to physical test observations when they become available.

The present trial is a controlled methodology demonstration, not a landing-impact simulation or a mass optimization. It uses a straight cantilever as a tractable benchmark. No physical testing observations are included, and the material cards are numerical assumptions. Consequently, the validation performed here compares model predictions with held-out FEA results generated using the same modeling framework; it does not establish agreement with a manufactured gear.

2 Finite-element model

The beam is a flat rectangular layered shell with fixed length $L = 0.25\,\mathrm{m}$. The axis is global X, width is global Y, and the applied transverse force is global $-Z$. The root edge is fully fixed. A total 10 N load is distributed equally across nodes on the free tip edge. The updated analysis is nonlinear static with large-deflection effects enabled using MAPDL `NLGEOM,ON` [1]. Automatic time stepping is enabled with 10 initial substeps, a 1–100 substep range, up to 50 equilibrium iterations per substep, and full Newton-Raphson iterations. This accounts for geometric nonlinearity from large rotations while retaining the example's orthotropic linear-elastic material cards.

The mesh uses SHELL181 layered shell elements, with 40 divisions along the length and 6 across the width. Three through-thickness integration points are assigned per ply. This is a two-dimensional shell discretization with a layered section, not a three-dimensional solid model. All cases use the same mesh.

Dataset A uses a symmetric twill/glass stack built by repeating carbon twill (T) and glass (G) in the half-stack and mirroring about the mid-plane. Carbon twill plies are 0.20 mm and glass plies are 0.10 mm. Dataset B's alternate-layup set uses a symmetric repeating carbon UD (U), carbon twill (T), and glass (G) half-stack, truncated to the needed half-ply count and mirrored; both carbon ply types are 0.20 mm and glass is 0.10 mm.

Table 1: Stack thicknesses used in the two layup families.

Ply count	Twill/glass total thickness	UD/twill/glass total thickness	Unit
8	1.2	1.4	mm
16	2.4	2.8	mm
24	3.6	4.0	mm
32	4.8	5.4	mm

Orthotropic elastic and shear properties are specified in the example configuration. For the transfer experiment, a carbon factor scales the elastic and shear moduli of carbon plies, and a glass factor scales those of glass plies; density, Poisson ratios, and ply thickness do not change when these factors vary. The cards and cured ply values are computational assumptions, not measured project material properties. For target Set 5 only, the resin-ratio proxy remixes the effective elastic and shear moduli using an assumed epoxy matrix modulus of 3.0 GPa and Poisson ratio 0.35. The configured baseline resin fraction is 50 vol

The response is tip compliance,

$$C = \frac{|\bar{u}_{z,\mathrm{tip}}|}{P}, \tag{1}$$

where $\bar{u}_{z,\mathrm{tip}}$ is the mean vertical tip-edge displacement and $P = 10\,\mathrm{N}$. Compliance has units of m/N. Under geometric nonlinearity, $C = |\delta|/P$ can depend on load; the model's valid load domain is therefore the fixed 10 N used consistently in both datasets. The case files also record effective section properties, mass, ply sequence, thickness, signed deflection, $|\delta|/L$, FEA wall time, and an analysis signature used to prevent incompatible results from being resumed.

2.1 Inputs, outputs, and supported domain

The configuration is a JSON file with SI units. Beam length, load, width, ply thickness, material moduli, and density must be positive and finite. The present training design uses a 0.25 m beam, 10 N load, widths 0.02–0.10 m, ply counts 8, 16, 24, and 32, and carbon/glass scale pairs $(1, 1)$, $(1, 2)$, $(2, 1)$, and $(2, 2)$. Target geometry cases use widths 0.015–0.15 m and even ply counts up to 100. Only symmetric `tw_glass` and `ud_tw_glass` stacks made of zero-degree plies are supported. The output is mean vertical tip displacement in metres and compliance in m/N; the CSV also records mass, calculated section stiffnesses, design identifiers, nonlinear-analysis metadata, and elapsed time.

Missing required configuration sections, unsupported stack names, odd or out-of-range ply counts, invalid material cards, nonpositive model predictors, or disabled large-deflection/time-stepping settings cause a clear exception before MAPDL is launched. The current model does not accept a load range, nonzero ply orientations, damage/failure models, or arbitrary material families. Changing load or the FEA definition requires a fresh analysis dataset and model fit; stale linear-static CSVs are rejected by the new case-signature check.

3 Experiment design and data

3.1 Dataset A: model training

Dataset A contains the full 5 by 4 by 4 factorial design: five widths, four ply counts, and four carbon/glass property-factor combinations. The combinations are $(1, 1)$, $(1, 2)$, $(2, 1)$, and $(2, 2)$.

- Widths: 0.02, 0.04, 0.06, 0.08, and 0.10 m.
- Ply counts: 8, 16, 24, and 32.
- Carbon/glass stiffness factor pairs: $(1, 1)$, $(1, 2)$, $(2, 1)$, and $(2, 2)$.

This gives 80 FEA responses. The training cases span the base twill/glass stack and the configured carbon/glass stiffness variation.

3.2 Dataset B: target sets

Dataset B contains five 20-case sets (100 cases total). Sets 1–3 and 5 use the Dataset A widths and ply counts; geometry extrapolation uses its separately defined levels.

Table 2: Dataset B target design.

Target set	Change from Dataset A
Material interpolation	Twill/glass layup; carbon/glass factors $(1.5, 1.5)$.
Material extrapolation	Twill/glass layup; factors $(2.25, 0.75)$.
Layup transfer	Symmetric UD/twill/glass layup; factors $(1, 1)$.
Geometry extrapolation	Twill/glass layup; widths 0.015, 0.04, 0.06, 0.08, 0.15 m and ply counts 8, 16, 32, 100; factors $(1, 1)$.
Material extrapolation v2	Twill/glass; carbon/glass factors $(2.25, 0.75)$; double the baseline resin:fibre volume ratio.

Set 5 deliberately changes a third material attribute that the correction model does not fit. Its target FEA cards use a two-times resin:fibre ratio, while nominal and calibrated surrogate features assume the baseline ratio. The two observations still fit only carbon and glass stiffness multipliers; the validation therefore measures whether those corrections absorb an unmodelled resin-rich shift, not whether the resin ratio is identified. The matrix modulus is based on typical cured tensile modulus in the Westlake EPIKOTE LR285 technical data sheet; the matrix Poisson ratio and the 50

The layup-transfer case is important to interpret correctly. Its FEA targets were generated with the material cards in the configuration at factors $(1, 1)$. Those same configured cards and the known alternate ply sequence were used to calculate nominal laminate section properties for prediction. Thus nominal predictions were not made with no material knowledge: this set tests transfer to a different layup using known nominal material cards. During calibration, the model was allowed to adjust the carbon and glass factors around that baseline using only response observations. The configured truth factors were not supplied to the calibration optimizer.

4 Models, calibration, and validation

4.1 Section-property predictors

For each case, the effective laminate bending rigidity per unit width D_{eff} is calculated from the laminate stiffness matrices. The transverse-shear rigidity per unit width is approximated as

$$S_{\text{eff}} = \sum_k G_{xz,k} t_k. \tag{2}$$

For a case of width b, the bending and corrected shear rigidities are $EI = bD_{\text{eff}}$ and $kGA = \kappa b S_{\text{eff}}$, with $\kappa = 5/6$. The mechanical basis terms are

$$x_b = \frac{L^3}{3EI}, \qquad x_s = \frac{L}{kGA}. \tag{3}$$

4.2 Model families

Three models are fit to Dataset A only:

1. **Mechanics-informed compliance regression:** nonnegative, no-intercept least squares in x_b and x_s, $\widehat{C} = \beta_b x_b + \beta_s x_s$.
2. **Quadratic response surface:** second-order polynomial in standardized logarithms of width, total thickness, D_{eff}, and S_{eff}, fit to log compliance with ridge regularization.
3. **Gaussian process:** Matérn 5/2 kernel plus a white-noise term, using the same log predictors and log compliance.

Five-fold cross-validation on Dataset A estimates within-training-domain predictive performance. Target FEA responses are not used to fit or select the Dataset A models. The mechanics-informed linear bending/shear basis is retained as a transparent baseline; it does not model large-rotation kinematics and is not an exact nonlinear beam solution.

4.3 Calibration and held-out validation

For each of the five target sets, three calibration repeats are performed. A pair of target cases is randomly selected for each repeat; the same pair is used across model families. For each fitted model, nonlinear least squares estimates two multipliers, q_C and q_G, by minimizing the squared log-compliance residuals at the selected cases:

$$(q_C^*, q_G^*) = \arg\min_{q_C, q_G} \sum_{i \in \mathcal{C}} \left[\log \widehat{C}_i(q_C, q_G) - \log C_{i,\text{FEA}}\right]^2, \qquad |\mathcal{C}| = 2. \tag{4}$$

Candidate factors are applied to the configured ply material cards, after which D_{eff}, S_{eff}, EI, and kGA are recalculated. The trained model coefficients or surrogate surface are held fixed. The compliance observations are the calibration data; calculated section properties are model inputs, not target observations.

The other 18 cases are held out for validation in each repeat. The calibration cases do not appear in the validation metrics. Reported measures include MAPE, maximum absolute percentage error, RMSE, and R^2. Calibration fits two unknown factors from the algebraic minimum of two observations, so a poorly chosen pair may produce non-identifiable or unstable factors. Jacobian rank and condition number are recorded. Runtime is measured separately for each FEA case, model fit, five-fold cross-validation, and target calibration/prediction; these timings are machine-specific and are saved with the outputs.

5 Historical results from the linear-static baseline

These tables report only the historical linear-static run: 80 target cases across the first four sets. They do not include the new resin-ratio Set 5 and are not results from the updated 'NLGEOM,ON' experiment. The 180-case nonlinear sweep must be run before reporting current validation scores or runtime comparisons.

5.1 Dataset A cross-validation

Table 3: Five-fold cross-validation scores on Dataset A.

Model	MAPE (%)	RMSE (m/N)	R^2
Mechanics-informed	0.1083	4.364×10^{-6}	0.9999994
Quadratic surface	0.0053	2.797×10^{-7}	0.999999998
Gaussian process	0.0195	2.171×10^{-6}	0.9999999

The quadratic surface had the lowest cross-validation error over the sampled Dataset A grid. The mechanics-informed model is more directly interpretable, while its error was higher. These results do not themselves establish target-set or physical accuracy.

5.2 Target-set held-out performance

Table 4 gives the mean MAPE across the three repeats, with standard deviation across repeats. Each MAPE is calculated over that repeat's 18 held-out cases.

Table 4: Held-out MAPE before and after calibration, mean ± standard deviation across repeats.

Target set	Model	Nominal MAPE (%)	Calibrated MAPE (%)
Material interpolation	Mechanics-informed	50.150 ± 0.008	0.125 ± 0.061
	Quadratic	50.000 ± 0.001	0.006 ± 0.001
	Gaussian process	49.974 ± 0.003	0.011 ± 0.002
Material extrapolation	Mechanics-informed	98.805 ± 0.409	1.007 ± 1.553
	Quadratic	98.619 ± 0.422	0.967 ± 1.656
	Gaussian process	98.586 ± 0.422	0.947 ± 1.622
Layup transfer	Mechanics-informed	0.067 ± 0.005	0.122 ± 0.078
	Quadratic	0.204 ± 0.002	0.066 ± 0.036
	Gaussian process	0.081 ± 0.002	0.060 ± 0.010
Geometry extrapolation	Mechanics-informed	0.254 ± 0.009	0.305 ± 0.155
	Quadratic	0.283 ± 0.032	0.359 ± 0.102
	Gaussian process	0.159 ± 0.021	0.246 ± 0.090

For the same-layup material-interpolation set, nominal predictions assume factors $(1, 1)$ while the target uses $(1.5, 1.5)$. Calibration recovers factors near the target and lowers held-out MAPE from about 50% to below 0.2% for all models. For material extrapolation, nominal MAPE is about 99%; informative calibration pairs again reduce it substantially. However, one random pair (case IDs 37 and 39) consists of two 32-ply cases at widths 0.06 and 0.10 m. Width changes both bending and shear compliance by a similar scale, so that pair does not separate carbon and glass effects. The mechanics Jacobian is rank deficient (rank 1; condition number about 2.2×10^{16}), and other models also produce poor scale estimates for that repeat. The three-repeat average therefore hides a calibration-design failure; the fitted factors and conditioning must be examined alongside prediction scores.

5.3 Dataset B subset 3: alternate-layup detail

Subset 3 is the 20-case `layup_transfer` set, IDs 40–59. Its widths are 0.02–0.10 m and ply counts are 8, 16, 24, and 32. All FEA target cases have configured material factors $(1, 1)$. Dataset A is the only training data; subset 3 is used solely for calibration and validation.

The corresponding validation IDs are: repeat 1, 40–47, 49–54, 56–59; repeat 2, 40–53, 55, 57–59; repeat 3, 40–53, 55–57, 59. Thus, each repeat validates on 18 cases and excludes only its own calibration pair. Fitted factors remain near $(1, 1)$, consistent with the configured target cards. Nominal predictions were already accurate because the model received those known cards and the alternate layup's calculated effective properties. Calibration modestly improved quadratic and Gaussian-process average errors but worsened the mechanics-informed average. This is not a test of estimating entirely unknown material properties: it is a layup-transfer test with known nominal material cards and optional response-based adjustment.

Table 5: Subset 3 calibration pairs, fitted scale factors, and held-out MAPE.

Model	Repeat	Calibration IDs	q_C	q_G	Nominal MAPE (%)	Calibrated MAPE (%)
Mechanics	1	48, 55	1.0022	0.9736	0.0719	0.0877
Mechanics	2	54, 56	1.0065	0.9195	0.0668	0.2112
Mechanics	3	54, 58	1.0003	1.0178	0.0625	0.0669
Quadratic	1	48, 55	0.9965	1.0241	0.2062	0.0505
Quadratic	2	54, 56	0.9946	1.0483	0.2046	0.1070
Quadratic	3	54, 58	0.9975	1.0000	0.2024	0.0395
Gaussian process	1	48, 55	0.9985	1.0082	0.0829	0.0481
Gaussian process	2	54, 56	0.9980	1.0110	0.0800	0.0648
Gaussian process	3	54, 58	0.9981	1.0071	0.0790	0.0659

6 Discussion and limitations

The results show that a small number of observations can correct a response model when the target variation is represented by the calibration parameters and the selected observations are informative. They also show that low Dataset A cross-validation error does not guarantee robust target calibration. Calibration pair selection matters: two response measurements should produce distinct sensitivities to the unknown property factors. A high Jacobian condition number or deficient rank signals that the fitted factors should not be interpreted.

The alternate-layup and geometry-extrapolation cases had low nominal MAPE because their configured section properties were available and close to the training response domain. Calibration brought little benefit there and sometimes increased error by adjusting the property factors to absorb small model residuals. This is consistent with a correction method that should be applied only when justified by observations and diagnostics.

The updated geometric-nonlinear analysis is intended to account for large rotations at the configured 10 N load. The prior linear-static run had a maximum target $|\delta|/L$ of 1.70 (21/80 target cases and 22/80 training cases exceeded 0.1); those old displacements and compliance values are not evidence for the new nonlinear formulation. Large rotations do not address material nonlinearity, laminate failure, progressive damage, contact, or landing-impact dynamics. Because compliance is load-dependent in this model, these predictions are restricted to the fixed 10 N case unless new load-level data are generated and included in training.

All cases are designed to use a 40 by 6 mesh. The previous result archive does not include a mesh-convergence study for this exact transfer design, and the updated nonlinear run has not yet supplied solver logs or runtime measurements. Before interpreting model scores, verify mesh sensitivity, reaction balance, element orientation, supports, load application, nonlinear convergence histories, and possible locking. The results do not address impact dynamics, failure, progressive damage, manufacturing constraints, or curved landing-gear geometry.

7 Conclusion and next steps

The prior linear-static trial implemented the training-calibration-validation workflow and compared three surrogate types. The updated runner now defines a nonlinear large-deflection static formulation, explicit input checks, case-signature protection against stale output reuse, and runtime instrumentation. The new formulation still requires a complete MAPDL rerun before its fit, calibration, validation and cost can be assessed. The prior trial showed that nominal predictions require a baseline material description, and calibration observations must identify the unknown factors. Its new-layup subset used known configured properties; it was not a no-material-knowledge test.

Next, generate all 180 cases using 'NLGEOM,ON', review solver convergence and force/reaction bal-

ance, then refit all three models and report new validation and timing results. Select calibration pairs by maximizing sensitivity separation and verify mesh convergence for representative cases. A subsequent study can introduce observations from a trusted analytical model or physical testing. Applying the method to curved gear geometry and optimizing mass/stroke are future extensions, not outcomes of this trial.

References

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